

random assignment (or before the time of eligibility determination for ENPs) and an eighteen month post-random assignment period.¹⁶

The ENP data we have collected to construct a non-experimental comparison group are very rich compared to the information available to previous analysts. Table 1 presents a description of the data used to construct non-experimental comparison groups in some major evaluations of job training programmes. The final column describes the JTPA data. None of the previous studies could accurately determine whether persons in the comparison groups were eligible for the programme being studied. Some lacked basic demographic information. Few had access to monthly data on earnings or on recent labour force histories. None located comparison groups members in the same labour markets as the participants, and because of the lack of geographical information, no adjustment for local labour market conditions was possible. Several studies—including the influential LaLonde study—used data that administered different questionnaires to participants and comparison group members with different definitions of earnings and reporting frames.

In light of the evidence presented below that geographical misalignment and misalignment of concepts used in questionnaires is a major source of evaluation bias, it is clear that previous analysts evaluating training programmes operated with their hands tied. In light of the evidence in Heckman and Smith (1994) that recent unemployment histories are important predictors of participation in job training programmes, it is significant that most of the major nonexperimental evaluations did not have access to this critical piece of information.

8. ESTIMATING THE PROBABILITY OF PARTICIPATION $P(X)$

We estimate logit models of programme participation for eligible persons. Predictor variables are chosen to maximize the within-sample correct prediction rates using the hit or miss method.¹⁷ For all four demographic groups analysed in this paper, recent unemployment histories are powerful predictors of participation. Recent earnings in the six months prior to entering the programme also increase the prediction rate, and are therefore included in the model. Appendix D and Appendix E, available on request, define the regressors used in this paper and present estimated logit coefficients.¹⁸

Figure 1 presents a major finding noted in our previous papers.¹⁹ For all four demographic groups, histograms of estimated probabilities of programme participation for ENPs ($D=0$) and controls ($D=1$) reveal that the estimated support common to both distributions of P is very limited. Since $B(X) = \tilde{B}(P(X))$ is only defined for values of P common to the supports of ENP and control group members, in general the bias is not defined over the full P support of either ENP or control group members. The parameter $M(S)$ estimated experimentally is not necessarily the same as $M(S)$ estimated non-experimentally if programme outcomes depend on the support. Not only are the supports different, but shapes of the distributions of P differ over the common support. We next

16. Additional detail about these surveys and about the construction of variables used in the analysis is presented in Appendix E, available on request from the authors.

17. The method classifies an observation as “1” if the estimated $P(X)$, satisfies $\hat{P}(X) > \hat{P}$, the sample proportion of eligible persons taking training and “0” otherwise. This method maximizes the overall classification rate for the sample assuming that the costs of misclassification are equal for the two groups. (See, e.g. Breiman, Friedman, Ohlsen and Stone, 1984).

18. Todd (1995) considers using semiparametric estimates of $P(X)$ and demonstrates that they have little effect on the estimates of bias presented in this paper. Therefore, for computational simplicity, we use the logit model. We use the weighting procedure first suggested by Rao (1965, 1986) and applied by Manski and Lerman (1977) to account for choice-based sampling.

19. Heckman and Roselius (1993) and Heckman, Ichimura, Smith and Todd (1996a, b).

TABLE 1
Comparison groups used in different studies

Study	Ashenfelter (1978)	Ashenfelter and Card (1985)	Dickinson-Johnson-West (1987)	Westat (1986) (Rupp and Bryant)
Programme, year, outcome variable	MDTA Classroom Trainees (First 3 months 1964), 1965-1969 annual social security record earnings	1976 CETA Trainees (1977, 1978 annual social security record earnings), CLMS data	1978 annual social security earnings. CETA trainees enrolled in 1976, CLMS data	2 cohorts of CETA Trainees 1977, 1978 annual social security record earnings, CLMS data
(1) Comparison group in the same labour market?	No	No	No	No
(2) Same questionnaire administered to comparison and treatment group	Yes	Yes	Yes	Yes
(3) Matching criteria (criteria for membership in comparison sample is also called "screening" criteria)	None specified	(a) 1975 earnings $\leq$ \$20K Household income $\leq$ \$30K (b) In labour force (March, 1976) Matched on age (persons $\geq$ 21 used)	(Matching based on a metric over vectors of variables) Matched on predictors of 1978 earnings including lagged earnings (1975-1970), worked in public sector, sex, and demographics. In labour force, March, 1976	Match on 1976 earnings, change in 1976 earnings (1975-1976, 1974-1975) change in earnings, demographics, 1975 labour force status, family income (for 1976-1977 cohort one year previous for 1975-1976 cohort). Either in the labour force, 1975 or at interview March 1976. Three matching groups based on income.
(4) Eligibility for programme known for comparison group members?	No	No	No	No
Variables used in analysis				
Age, race, sex	Yes (No age restriction)	Yes (Age $\geq$ 21 years old)	Yes (Age 21-65)	Yes (Age 14-60)
Education	No	Yes	Yes	Yes
Training history	No	No	No	No
Children	No	No	No	No
Employment histories	No	No	Yes (recent)	Yes (recent)
Hours worked	No	Yes	Yes	Yes
Unemployment histories	No	No	Yes (recent)	Yes (recent)
On welfare	No	No	Yes	No
Earning histories**	(Annual earnings) 5 years pre-programme 5 years post-programme	(Annual earnings) 2 years pre-programme 2 years post-programme	Same as Ashenfelter and Card (1985)	(Annual earnings) 4 years pre-enrollment earnings histories

** CLMS data matched Social Security Longitudinal Records to March CPS data for 1976 and 1977. The CPS data are for comparison group members. SSA data on longitudinal earnings are available for both groups. All of the personal and family information available in the CPS including short-term employment and labour-force participation histories are available but not necessarily used in the analysis. The CLMS studies all use the social security earnings data.

Study	NSW (supported work) data			JTPA data
	LaLonde (1986)	Fraker and Maynard (1987) and LaLonde and Maynard (1986)		
Programme, year, outcome variable	Annual earnings 1978 annual social security earnings and PSID earnings NSW (Supported Work) Data	1977, 1978, 1979 annual earnings for AFDC recipients and for youth NSW (Supported Work) Data	Quarterly and monthly earnings 1987-1989	
(1) Comparison group in the same local labour market	No	No	Yes	
(2) Same questionnaire administered to comparison and treatment group?	No	No	Yes	
(3) Matching criteria (criteria for membership in comparison sample is also called "screening criteria")	PSID: (a) Men and Women who are household heads 1975-1979 CPS: Matches March 1976 CPS earnings with SSA earnings. Person with 1976 income $\leq$ \$20K and household income $\leq$ \$30K	Three Samples (i) Eligible in sample period: for youth: high school dropout-exclude in school youth. For AFDC: age of youngest child, receipt of welfare matching. (ii) Cell matching: based on predictors of 1979 SSA earnings of eligibles: (earnings prior to programme participation), demographics, education, family income, change in earnings. (iii) Stratified matches on imputed 1979 earnings: earnings estimated on eligible nonparticipant sample plus demographic criteria (race, sex). Same criteria for prediction as in (ii).	Persons screened to be eligible for JTPA; out of school youth, no disabled persons; Title IIA only	
(4) Eligibility for programme known for comparison group members?	No	No	Yes	
Variables used in analysis				
Age, race, sex	Yes: Women AFDC recipients 20-55, Males $\leq$ 55	Same as LaLonde	Yes	
Education	Yes	Same as LaLonde	Yes	
Training history	No	Same as LaLonde	Yes	
Children	Yes	Same as LaLonde	Yes	
Employment histories	No	Same as LaLonde	Yes	
Hours worked	No	Same as LaLonde	Yes	
Unemployment histories	No	Same as LaLonde	Yes	
Welfare receipt?	Yes	Same as LaLonde	Yes	
Earnings histories	Two years post-programme Two years pre-programme	Same as LaLonde	(Five years of pre-programme earnings) monthly earnings	

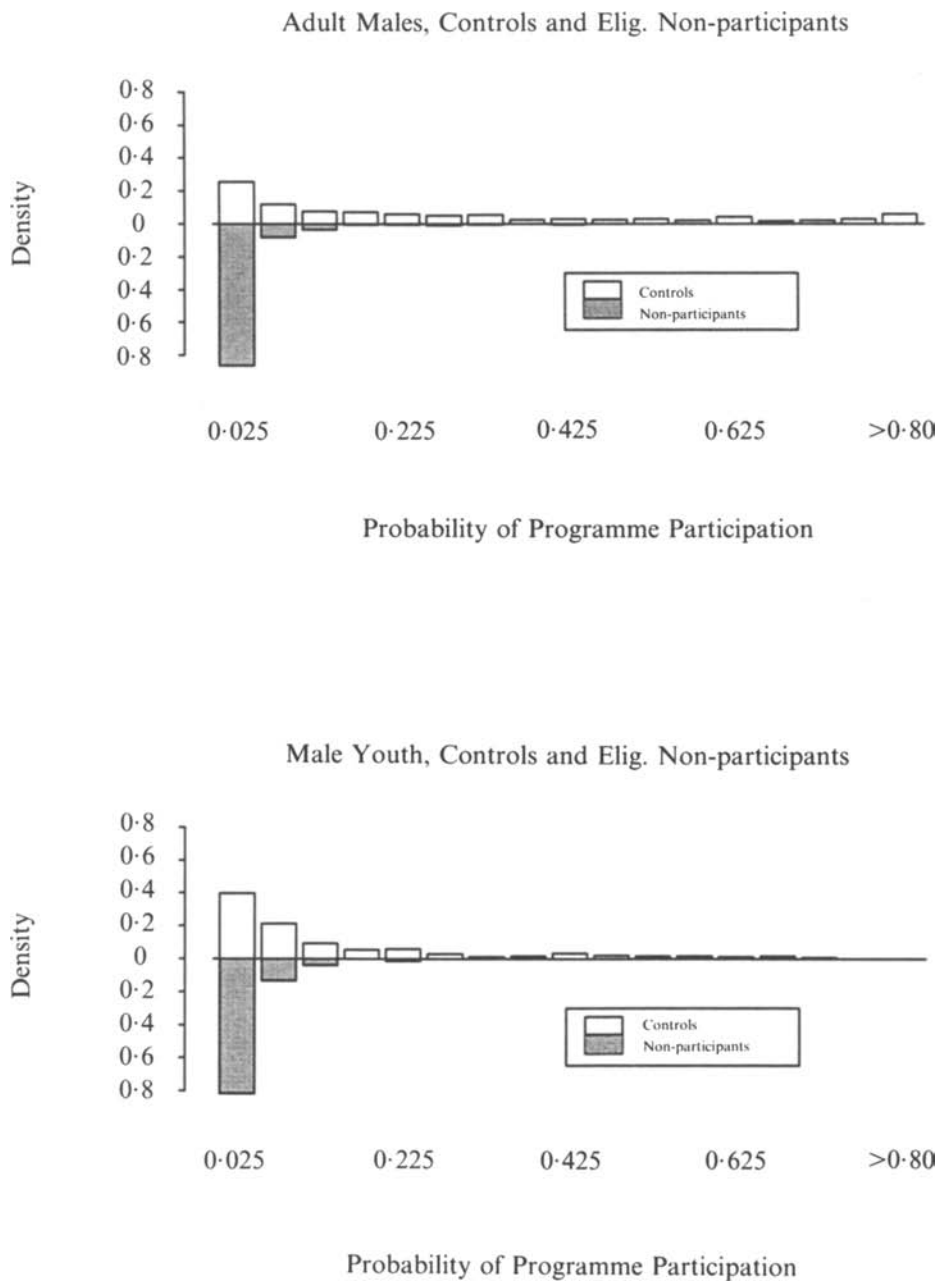


FIGURE 1. Density of Estimated Probability of Programme Participation